

YOHANNES NIGUSSE

Software Engineer Intern | Backend & Systems

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EDUCATION

St. Cloud State University | St. Cloud, MN

B.S. Computer Science, Artificial Intelligence & Machine Learning Track | Expected December 2027

GPA: 3.92 | Coursework: Distributed Systems, Relational Databases & SQL, Operating Systems, Software Design & Architecture, Object-Oriented Programming, Data Structures & Algorithms

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Java, C#, Bash

Backend & APIs: FastAPI, Flask, REST API, REST APIs, microservices, backend services, internal tools, data pipelines, relational databases (PostgreSQL, SQLite), pgvector, MongoDB, NoSQL, SQLAlchemy, query optimization, data systems

Cloud & Systems: Distributed systems, scalable architectures, scalability, reliable systems, reliability, cloud services (GCP, AWS Lambda/API Gateway, Azure Storage/Search), Docker, Linux/Ubuntu, Git, GitHub Actions, CI/CD, production deployment

Testing & SDLC: Unit testing, integration testing, regression testing, code review, debugging, troubleshooting, validation, documentation, software design, object-oriented programming

AI & Modern Dev: Machine learning, LLM pipelines, RAG, AI-assisted development (Claude Code, GitHub Copilot), model evaluation, OpenAI API, Gemini API

Operations & Methods: Agile work tracking, collaboration, operational monitoring, logs, telemetry, observability, security hardening, enterprise software, ownership

PROFESSIONAL EXPERIENCE

Accenture | Seattle, WA

Technology Summer Analyst | June 2026 - August 2026

- Resolved 10+ externally reported accessibility defects in programmatically generated PowerPoint decks, remediating source-level issues in a release-critical platform.
- Engineered a custom accessibility layer in the TypeScript/OOXML generation pipeline and authored 11 Jest regression rules for screen-reader compatibility, preventing accessibility regressions before production deployment.
- Validated accessibility fixes across 8 formal scenarios with NVDA, covering reading order, alt text, title structure, navigation, footnotes, and malformed bullets; delivered sign-off evidence clearing production release for ~30,000 internal accounts.

Kibur College | Remote

Software Engineering Intern | January 2026 - May 2026

- Built and owned a multi-channel RAG admissions assistant for Telegram and email as sole developer, implementing dense retrieval and Gemini LLM synthesis to achieve 93.3% correct-or-partially-correct responses across 84 queries and 12 active users.
- Designed a 75-question retrieval benchmark validating language model context grounding, achieving 90.7% expected-context retrieval and using 7 isolated failed-context cases to guide pipeline tuning.
- Co-designed a shared-nothing distributed backend across 60+ repurposed campus computers, modeling scalable capacity planning, quorum-based availability, and fault-isolated horizontal scaling for admissions workloads without cloud infrastructure spend.
- Built a SQLite-backed staff dashboard for approximately 12 staff members, enabling non-technical institutional knowledge updates, decoupling content changes from code deployments, and eliminating developer bottlenecks.

Brain-Computer Interface Lab, St. Cloud State University | St. Cloud, MN

Undergraduate Research Assistant | August 2024 - May 2025

- Diagnosed and resolved an EEG data pipeline failure caused by sample-rate dimension mismatches, implementing automated resampling that restored 14 affected recordings across 3 data-collection runs.
- Hardened research infrastructure for 4 researchers by implementing RSA key-management scripts and resolving operational Linux/Ubuntu configuration issues across 2 workstations and 1 storage server, securing access controls for OpenBCI EEG workflows.

Center for Women in Technology, UMBC | Baltimore, MD

Technical Operations Assistant | September 2023 - May 2024

- Built Python and Google Sheets API automation for recurring event-attendance workflows serving 50-100 participants per session, reducing manual record processing by approximately 85%.

PROJECTS

NimbusQueue | Backend Lead, 4-person team

CSCI 312: Fundamentals of Distributed Systems | Python, FastAPI, PostgreSQL, Docker, GCP, GitHub Actions

- Load-tested a scalable FastAPI task-queue microservice with 5,000 mixed-duration jobs, reducing queue drain time 62% and improving P95 processing latency from 450ms to 110ms across a 3-worker GCP cluster.
- Validated failover by terminating the active coordinator during a 2,500-job workload; PostgreSQL-backed leader election restored processing in 3.4 seconds with 0 duplicated or lost jobs.
- Containerized the multi-worker environment with Docker Compose and configured GitHub Actions CI/CD to gate redeployments on a 42-case integration test suite, preventing regressions before release.

The Stack | Solo Developer

Python, PostgreSQL, OpenAI API, Docker, Streamlit

- Engineered a reliable data ingestion pipeline for YouTube and iCal feeds with run tracking, stage metrics, and item-level error isolation, powering personalized event and opportunity email digests without failed items halting a run.
- Designed and executed a 500-run reliability test on a reproducible input set, achieving 96.4% successful runs; built operational monitoring commands and structured JSON logging to trace degraded/failure cases to API rate limits.

ConcurrentBank | Solo Developer

CSCI 411: Database Theory & Design | PostgreSQL, Python, SQLAlchemy

- Stress-tested 1,500 concurrent financial transactions, reproducing a non-repeatable read under PostgreSQL Read Committed and verifying 0 occurrences under Repeatable Read post-change runs.
- Built a deterministic deadlock test that reproduced PostgreSQL 40P01 across 50 runs, then enforced consistent lock ordering and automated SQL query retries with 0 deadlocks across 50 post-change runs.
- Validated WAL crash recovery during a 500-transaction workload, confirming 312 committed transactions remained durable and 188 interrupted transactions stayed uncommitted.

LEADERSHIP & HONORS

Vice President, National Society of Black Engineers | ColorStack Affiliate | Student Government Technology Fee Committee | Dean's List | Monte Johnson CS Scholarship | Duane Joyer Memorial Scholarship